

INPPL1 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes INPPL1 gene fusions from the msk_impact_50k_2026 study, covering 14 fusion events. 4/14 fusions (28.6%) were in-frame. 7/14 fusions (50.0%) retained the Endonuclease/Exonuclease/phosphatase family domain. Domain retention was tested with Fisher's exact test ($p=0.048951$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for INPPL1.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 14 analyzed fusion events, identifying 11 distinct fusion partner genes. The most frequent partner was FCHSD2, observed in 4 events.

Domain retention

Endonuclease/Exonuclease/phosphatase family domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 4/4 (100.0%) of in-frame fusions retained the domain, $p=0.048951$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.108911.

Domain disruption

Disruption of the SH2 domain was tested with Fisher's exact test comparing in-frame fusions against all others; 0/4 (0.0%) of in-frame fusions disrupted the domain, $p=1$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 1.

Cutpoint detection

Cutpoint detection scanned 13 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 560 aa (permutation-corrected $p=0.033966$, statistically significant at $\alpha=0.05$). This is 131 aa from the nearest configured domain boundary at 429 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=7$) and "not_retained" ($n=6$). For Frame_status, retained: 4/5 (80.0%, 95% CI 37.6%-96.4%); not_retained: 0/2 (0.0%, 95% CI 0.0%-65.8%). Welch's t-test comparing Tumor_variant_count between groups gave means of 19.7143 vs 43.3333, $p=0.434221$ (not statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score produced results for this run (11 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Joint partner

Joint partner reported: INPPL1 has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ANK2	1	0.07142857142857142	0.09629286886244175	0.0	1.0	0.37706948295080434	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2332087369392623	1
SHANK2	1	0.07142857142857142	0.09629286886244175	0.0	0.956	0.37706948295080434	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2266087369392623	2
FCHSD2	4	0.2857142857142857	0.09629286886244175	0.0	0.51	0.37706948295080434	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.22399445122497658	3
PTCH1	1	0.07142857142857142	0.09629286886244175	0.0	0.608	0.37706948295080434	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1744087369392623	4
GAB2	1	0.07142857142857142	0.09629286886244175	0.0	0.31999999999999995	0.37706948295080434	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.13120873693926227	5
NLRP6	1	0.07142857142857142	0.09629286886244175	0.0	0.29200000000000004	0.37706948295080434	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1270087369392623	6
CLPB	1	0.07142857142857142	0.09629286886244175	0.0	0.246	0.37706948295080434	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12010873693926229	7
FOLR2	1	0.07142857142857142	0.09629286886244175	0.0	0.14600000000000002	0.37706948295080434	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1051087369392623	8
FAT3	1	0.07142857142857142	0.09629286886244175	0.0	0.10799999999999998	0.37706948295080434	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.09940873693926229	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
SLC67A1	1	0.07142857142857142	0.09629286886244175	0.0	None	0.37706948295080434	['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']	0.09789263169324977	10
LRTOMT	1	0.07142857142857142	0.09629286886244175	0.0	0.0	0.37706948295080434	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.0832087369392623	11

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
114	0	7	1	5	0.0	0.46153846153846156	0.3357921019231931
133	0	7	2	4	0.0	0.1923076923076923	0.7160033436347992
183	1	6	2	4	0.3333333333333333	0.5594405594405595	0.2522460504731182
206	1	6	3	3	0.16666666666666666	0.26573426573426573	0.5755524408482516
220	1	6	4	2	0.08333333333333333	0.10256410256410256	0.9890046156985368
364	2	5	4	2	0.2	0.2861305361305362	0.5434357913897181
538	2	5	5	1	0.08	0.10256410256410256	0.9890046156985368
560	2	5	6	0	0.0	0.020979020979020976	1.6782147827453995
805	6	1	6	0	0.0	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

0	3
4	6

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf00017	SH2 domain	13	10	0	3	0.0

domain_retention tables

frame_domain_contingency_table

4	3
0	6

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	Endonuclease/Exonuclease/phosphatase family	13	7	2	4	0.17988505747126438

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
11	8	14	0.5714285714285714

frequency tables

Partner_gene_counts

Partner_gene	Event_count
ANK2	1
CLPB	1
FAT3	1
FCHSD2	4
FOLR2	1
GAB2	1
LRTOMT	1
NLRP6	1
PTCH1	1
SHANK2	1
SLC67A1	1

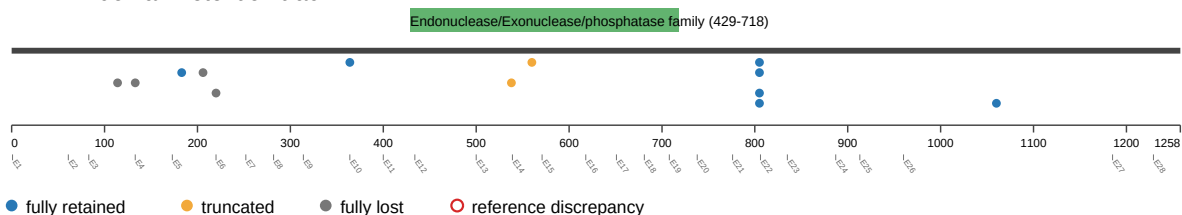
Per-event results (results.tsv)

event_id	samp le_id	p a t i e n t _ i d	fusio n_n ame	par tne r_g ene	fra me _st atu s	ta rg et _r ole	break point_ geno mic	brea kpoi nt_e xon	breakpoi nt_protei n_positio n	is_intr onic_b reakpoi nt	do mai n_s tatu s	domain_ retained_ _fraction	domai n_is_t runcat ed	retained_dom ains	lost_domains	disrupted_do mains	source_annotation_te xt	source_sit e2_effect_ on_frame
EVT-P-0 007343-T 02-IM6-4	P-000 7343- T02-I M6		INP PL1- ANK 2	AN K2	un kn ow n	fiv e_ pri me	71943 348	14	560	False	disr upt ed	0.455172 4137931 0347	True	SH2 domain		Endonuclease/ Exonuclease/p hosphatase family	INPPL1 (NM_001567) - ANK2 (NM_001148) rearrangement: t(4;11)(q25-q26;q13.4)(chr4:g.1 13898703::chr11:g.719 43348) Protein Fusion: mid-exon {INPPL1:ANK2}	NA
EVT-P-0 031523-T 02-IM6-5	P-003 1523- T02-I M6		INP PL1- CLP B	CL PB	un kn ow n	thr ee_ _p ri me	71940 164	5	183	False	reta ined	1.0	False	Endonuclease/ Exonuclease/p hosphatase family	SH2 domain		INPPL1 (NM_00156) rearrangement: c.549:1 NPPL1_chr11:g.720245 01del Antisense Fusion	NA
EVT-P-0 042110-T 02-IM7-6	P-004 2110- T02-I M7		INP PL1- FAT 3	FA T3	un kn ow n	fiv e_ pri me	71939 487	3	114	False	lost	0.0	False	SH2 domain	Endonuclease/ Exonuclease/p hosphatase family		INPPL1 (NM_001567) - FAT3 (NM_001008781) rearrangement: c.342:1 NPPL1_c.3669+26772: FAT3del Protein Fusion: mid-exon {INPPL1:FAT3}	NA
EVT-P-0 004226-T 08-IM7-7	P-000 4226- T08-I M7		INP PL1- FCH SD2	FC HS D2	in-f ra me	fiv e_ pri me	71945 806	21	805	True	reta ined	1.0	False	Endonuclease/ Exonuclease/p hosphatase family; SH2 domain			INPPL1 (NM_001567) - FCHSD2 (NM_014824) rearrangement: c.2415+ 147:INPPL1_c.705+164 42:FCHSD2inv Protein Fusion: in frame {INPPL1:FCHSD2}	NA
EVT-P-0 004226-T 04-IM6-8	P-000 4226- T04-I M6		INP PL1- FCH SD2	FC HS D2	in-f ra me	fiv e_ pri me	71945 806	21	805	True	reta ined	1.0	False	Endonuclease/ Exonuclease/p hosphatase family; SH2 domain			INPPL1 (NM_001567) - FCHSD2 (NM_014824) rearrangement: c.2415+ 147:INPPL1_c.705+164 42:FCHSD2inv Protein Fusion: in frame {INPPL1:FCHSD2}	NA
EVT-P-0 004226-T 03-IM6-9	P-000 4226- T03-I M6		INP PL1- FCH SD2	FC HS D2	in-f ra me	fiv e_ pri me	71945 742	21	805	True	reta ined	1.0	False	Endonuclease/ Exonuclease/p hosphatase family; SH2 domain			INPPL1 (NM_001567) - FCHSD2 (NM_014824) rearrangement: c.2415+ 83:INPPL1_c.705+1644 0:FCHSD2inv Protein Fusion: in frame {INPPL1:FCHSD2}	NA
EVT-P-0 004226-T 07-IM7-1 0	P-000 4226- T07-I M7		INP PL1- FCH SD2	FC HS D2	in-f ra me	fiv e_ pri me	71945 792	21	805	True	reta ined	1.0	False	Endonuclease/ Exonuclease/p hosphatase family; SH2 domain			INPPL1 (NM_001567) - FCHSD2 (NM_014824) rearrangement: c.2415+ 133:INPPL1_c.705+164 41:FCHSD2inv Protein Fusion: in frame {INPPL1:FCHSD2}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0045578-T01-IM6-11	P-0045578-T01-IM6		INPPL1-FOLR2	FOLR2	out-of-frame	five_prime	71939641	3	133	True	lost	0.0	False	SH2 domain	Endonuclease/Exonuclease/p hosphatase family		INPPL1 (NM_001567) - FOLR2 (NM_001113536) rearrangement: c.397+99:INPPL1_c.151-191:FOLR2dup Protein Fusion: out of frame {INPPL1:FOLR2}	NA
EVT-P-0009791-T02-IM6-12	P-0009791-T02-IM6		INPPL1-GAB2	GAB2	out-of-frame	five_prime	71940493	6	220	True	lost	0.0	False	SH2 domain	Endonuclease/Exonuclease/p hosphatase family		INPPL1 (NM_001567) - GAB2 (NM_080491) rearrangement: c.660-16:INPPL1_c.75+18325:GAB2inv Protein Fusion: out of frame {INPPL1:GAB2}	NA
EVT-P-0048151-T01-IM6-31	P-0048151-T01-IM6		INPPL1-LRTO MT	LRTOMT	unknown	five_prime	71948466	26	1060	False	retained	1.0	False	Endonuclease/Exonuclease/p hosphatase family; SH2 domain			INPPL1 (NM_001567) - LRTOMT (NM_145309) rearrangement: c.3178:INPPL1_c.83-1241:LRTOMTdup Protein Fusion: mid-exon {INPPL1:LRTOMT}	NA
EVT-P-0005693-T03-IM6-34	P-0005693-T03-IM6		INPPL1-PTCH1	PTCH1	out-of-frame	three_prime	71941318	9	364	True	retained	1.0	False	Endonuclease/Exonuclease/p hosphatase family	SH2 domain		PTCH1 (NM_000264) - INPPL1 (NM_001567) rearrangement: t(9;11)(q22.32;q13.4)(chr9:g.98211830::chr11:g.71941318) Protein Fusion: out of frame {PTCH1:INPPL1}	NA
EVT-P-0024858-T01-IM6-36	P-0024858-T01-IM6		NLRP6-INPPL1	NLRP6	unknown	five_prime	71940232	5	206	False	lost	0.0	False	SH2 domain	Endonuclease/Exonuclease/p hosphatase family		INPPL1 (NM_001567) - NLRP6 (NM_138329) rearrangement: c.617:INPPL1_c.2138:NLRP6dup Protein Fusion: mid-exon {INPPL1:NLRP6}	NA
EVT-P-0031882-T01-IM6-37	P-0031882-T01-IM6		SHANK2-INPPL1	SHANK2	unknown	three_prime	71942656	13	538	False	disrupted	0.6241379310344828	True		SH2 domain	Endonuclease/Exonuclease/p hosphatase family	INPPL1 (NM_001567) rearrangement: c.1612:INPPL1_chr11:g.70501515dup Antisense Fusion	NA

domain retention outliers

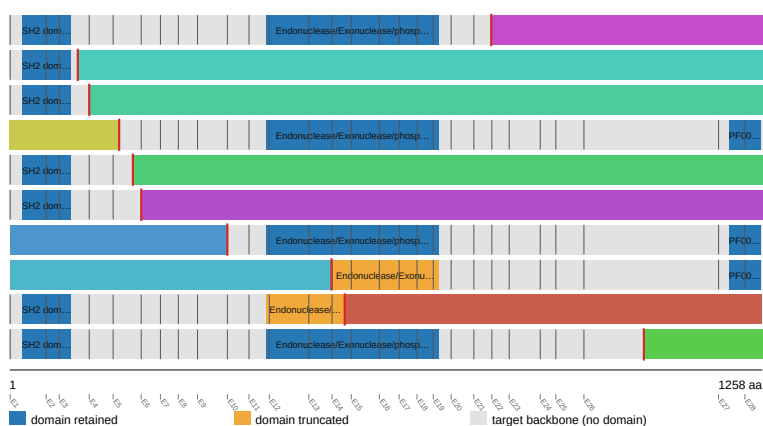
INPPL1 domain-retention track



fusion schematic

INPPL1 fusion-transcript schematic

partner block → breakpoint → retained INPPL1 portion (domain-colored, exon ticks)



FCHSD2 (x4) – Endonuclease/Exonuclease/phosphatase family retained

FAT3 – Endonuclease/Exonuclease/phosphatase family lost

FOLR2 – Endonuclease/Exonuclease/phosphatase family lost

CLPB – Endonuclease/Exonuclease/phosphatase family retained

NLRP6 – Endonuclease/Exonuclease/phosphatase family lost

GAB2 – Endonuclease/Exonuclease/phosphatase family lost

PTCH1 – Endonuclease/Exonuclease/phosphatase family retained

SHANK2 – Endonuclease/Exonuclease/phosphatase family truncated

ANK2 – Endonuclease/Exonuclease/phosphatase family truncated

LRTOMT – Endonuclease/Exonuclease/phosphatase family retained

■ breakpoint

reference comparison

Reference vs msk impact 50k 2026

